1 Assignment 1

1. What is the time for light to cross a proton?

   Solution:

   considering the diameter of proton $d = 1.6 \times 10^{-15}$ m

$$\begin{aligned} t &= \frac{d}{c} \\ &= \frac{1.6 \times 10^{-15}}{3 \times 10^{8}} \\ &= 5.33 \times 10^{-24}\text{s} \end{aligned}$$

2. Using the property of the metric tensor, show that:

$$A \cdot B = A^0 B^0 - \vec{A} \cdot \vec{B}$$

   Solution:

$$\begin{aligned} A.B &= g_{\mu\nu} A^{\mu} B^{\nu} \\ &= A^0 B^0 - \delta_{ij} A^i B^j \\ &= A^0 B^0 - \vec{A}.\vec{B} \end{aligned}$$

3. Show that:

$$\Lambda^{\alpha}{}_{\mu} \Lambda^{\beta}{}_{\nu} \eta_{\alpha\beta} = \eta_{\mu\nu}$$

   is a necessary condition for a Lorentz transformation.

   Solution:

   In Lorentz transformation $ds^2 = \eta_{\mu\nu} dx^{\mu} dx^{\nu}$ is invariant hence

$$\begin{aligned} \eta_{\mu\nu} dx^{\mu} dx^{\nu} &= \eta_{\mu\nu} dx'^{\mu} dx'^{\nu} \\ &= \eta_{\mu\nu} \Lambda^{\mu}_{\sigma} \Lambda^{\nu}_{\rho} dx^{\sigma} dx^{\rho} \end{aligned}$$

   and therefore

$$\boxed{\eta_{\mu\nu} = \eta_{\rho\sigma} \Lambda^{\rho}_{\mu} \Lambda^{\sigma}_{\nu}}$$

4. Relativistic wave equations contain derivatives with respect to space-time. How does a derivative with respect to space-time transform under a Lorentz transformation?

   Solution:

The covariant vector transforms according to

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$$

in Lorentz transformation
derivative with respect to spacetime ∂_μ is a covariant vector so they transform as

$$\partial'_\mu = \Lambda_\mu{}^\nu \partial_\nu$$

where $\Lambda_\mu{}^\nu = (\Lambda^\mu{}_\nu)^{-1}$

5. A particle of mass M, initially at rest, decays into two pieces, each of mass m. What is the speed of each particle?
   Solution:
   Using momentum conservation we'll have momenta of both particles to be same
   Now energy conservation will give

$$\begin{aligned} Mc^2 &= 2\gamma mc^2 \\ \frac{1}{\sqrt{1-v^2/c^2}} &= \frac{M}{2m} \\ \frac{v^2}{c^2} &= 1 - \frac{4m^2}{M^2} \end{aligned}$$

$$\boxed{v = c\sqrt{1 - \frac{4m^2}{M^2}}}$$

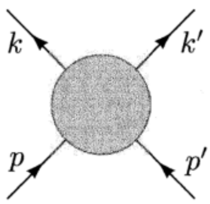
6. Consider the decay $X \to a + b$ where b is massless. What is the speed of a?
   Solution:
   Let mass of X be M and that of a be m. Assume particle X initially at rest, therefore both a and b will have same momentum $\mathbf{p}$

Using Energy Conservation

$$
\begin{aligned}
M &= \sqrt{p^2 + m^2} + p \\
(M - p)^2 &= p^2 + m^2 \\
M^2 - 2Mp &= m^2 \\
2M\gamma m v &= M^2 - m^2 \\
\frac{v^2}{1 - v^2} &= \left(\frac{M^2 - m^2}{2Mm}\right)^2 \\
v &= \frac{\frac{M^2 - m^2}{2Mm}}{\sqrt{1 + \left(\frac{M^2 - m^2}{2Mm}\right)^2}} \\
v &= \frac{M^2 - m^2}{\sqrt{4M^2m^2 + (M^2 - m^2)^2}}
\end{aligned}
$$

$$\boxed{v = \frac{M^2 - m^2}{M^2 + m^2}}$$

7. Show that the Mandelstam variables s, t, u are invariant.



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$$
\begin{aligned}
s &= (p + p')^2 \\
&= (p + p')_\mu (p + p')^\mu \\
&= p^\mu p_\mu + p'^\mu p'_\mu + 2p^\mu p'_\mu \\
&= m_p^2 + m_{p'}^2 + 2p^\mu p'_\mu
\end{aligned}
$$

All $m_p^2, m_{p'}^2$ and $p^\mu p'_\mu$ are Lorentz Scalars, hence are invariant. Thus s is Lorentz invariant

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$$
\begin{aligned}
t &= (k - p)^2 \\
&= m_k^2 + m_p^2 - 2k^\mu p_\mu
\end{aligned}
$$

Hence Lorentz Invariant

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$$\begin{aligned} u &= (k' - p)^2 \\ &= m_{k'}^2 + m_p^2 - 2k'^{\mu} p_{\mu} \end{aligned}$$

Again Lorentz Invariant

8. Show that in a $2 \to 2$ scattering process:

$$s + t + u = \sum_i m_i^2$$

Solution:

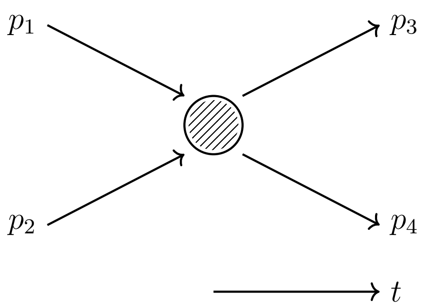

Figure 1:

$$\begin{aligned} s &= (p_1 + p_2)^2 = (p_3 + p_4)^2 \\ t &= (p_3 - p_1)^2 = (p_4 - p_2)^2 \\ u &= (p_4 - p_1)^2 = (p_3 - p_2)^2 \end{aligned}$$

$$\begin{aligned} s + t + u &= p_1^2 + p_2^2 + p_3^2 + p_1^2 + p_4^2 + p_1^2 + 2p_1.p_2 - 2p_1.p_3 - 2p_1.p_4 \\ &= 3p_1^2 + p_2^2 + p_3^2 + p_4^2 + 2p_1.(p_2 - p_3 - p_4) \end{aligned}$$

now by conservation of four momentum

$$\begin{aligned} p_1 + p_2 &= p_3 + p_4 \\ p_2 - p_3 - p_4 &= -p_1 \end{aligned}$$

therefore,

$$\begin{aligned} s+t+u &= 3p_1^2 + p_2^2 + p_3^2 + p_4^2 - 2p_1^2 \\ &= p_1^2 + p_2^2 + p_3^2 + p_4^2 \end{aligned}$$

$$\boxed{s+t+u = \sum_i m_i^2}$$

9. Show that when all masses are neglected in a $2 \to 2$ reaction, then:

$$t = -\frac{s}{2}(1-\cos\theta), \qquad u = -\frac{s}{2}(1+\cos\theta)$$

where θ is the scattering angle.

Solution:

Scattering Angle is angle between incoming particle and outgoing particle in center of mass frame

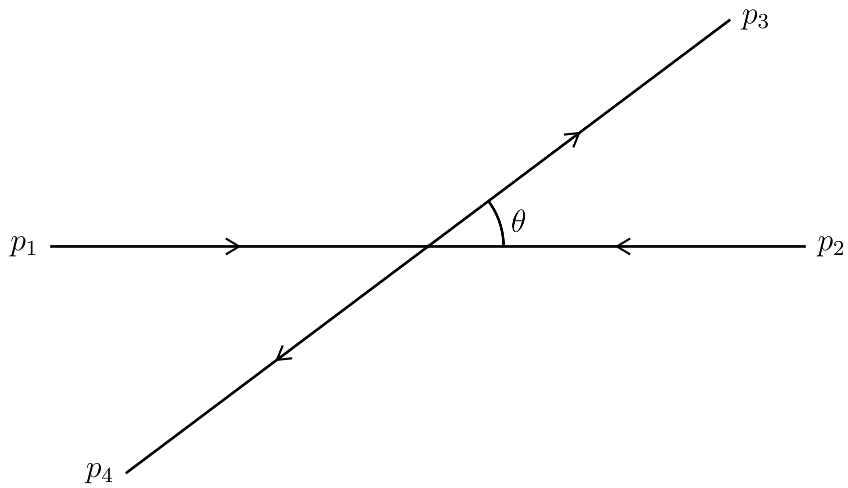

Figure 2: 2-2 Scattering in Center of Mass Frame

$$\begin{aligned} p_1 &= (E, E, 0, 0, 0) \\ p_2 &= (E, -E, 0, 0) \\ p_3 &= (E, E\cos\theta, E\sin\theta, 0) \\ p_4 &= (E, -E\cos\theta, -E\sin\theta, 0) \end{aligned}$$

Now,

$$\begin{aligned}
t &= (p_1 - p_3)^2 \\
&= -2p_1.p_2 \quad (\text{since } m_i = 0) \\
&= -2(E^2 - E^2\cos\theta) \\
&= -2E^2(1-\cos\theta)
\end{aligned}$$

Similarly,

$$\begin{aligned}
s &= (p_1 + p_2)^2 \\
&= 4E^2 \\
E^2 &= \frac{s}{4}
\end{aligned}$$

Hence,

$$\boxed{t = -\frac{s}{4}(1-\cos\theta)}$$

In similar way we can show

$$\begin{aligned}
u &= (p_1 - p_4)^2 \\
&= -2p_1.p_4 \\
&= -2E^2(1+\cos\theta)
\end{aligned}$$

$$\boxed{u = -\frac{s}{4}(1+\cos\theta)}$$

10. A particle of mass M, initially at rest, decays into two particles, each of mass m. What is the speed of each as they fly off?
 Solution:
 Same question as question 5

11. Show that in the rest frame of a decaying particle $a \to b + c$:

$$E_b = \frac{M_a^2 - M_c^2 + M_b^2}{2M_a}$$

 Solution:
 we write four-momentum of particles as

$$p_a = (M_a, \mathbf{0}) \qquad p_b = (E_b, \mathbf{p}) \qquad p_c = (E_c, -\mathbf{p})$$

 also,

$$E_b^2 = \mathbf{p}^2 + M_b^2 \tag{1}$$

$$M_a = E_b + E_c \tag{2}$$

Now

$$\begin{aligned}
p_a^2 &= p_b^2 + p_c^2 + 2p_a.p_b \\
M_a^2 &= M_b^2 + M_c^2 + 2(E_bE_c + \mathbf{p}^2) \\
&= M_b^2 + M_c^2 + 2(E_b(M_a - E_b) + E_b^2 - M_b^2) \qquad \text{(using (1) and (2))} \\
&= -M_b^2 + M_c^2 + 2M_aE_b
\end{aligned}$$

$$\boxed{E_b = \frac{M_a^2 - M_c^2 + M_b^2}{2M_a}} \tag{3}$$

12. Suppose two identical particles, each of mass m and kinetic energy T, collide head-on. Find the kinetic energy T' in the fixed-target frame of reference.
    Solution:
    Let particles collide head on. To find T' consider the transformation of E in Lorentz transformation with boost such that one particle is at rest

$$E' = \gamma(E - p^1v^1)$$

We also have relations

$$p = \sqrt{T^2 + 2Tm}$$

$$v = \frac{p}{E} = \frac{\sqrt{T^2 + 2Tm}}{T + m}$$

Therefore,

$$\begin{aligned}
T' + m &= \frac{T + m + \sqrt{T^2 + 2Tm} \times \frac{\sqrt{T^2+2Tm}}{T+m}}{\sqrt{1 - \frac{T^2+2Tm}{(T+m)^2}}} \\
&= \frac{(T+m)^2 + T^2 + 2Tm}{m} \\
&= \frac{2T^2 + 4Tm}{m}
\end{aligned}$$

$$\boxed{T' = \frac{2T(T + 2m)}{m}}$$

13. Consider the decay $\pi^+ \to \mu^+\nu_\mu$, where $E_{\pi^+} = 20$ GeV and $\tau_{\pi^+} = 2.6 \times 10^{-8}$ s. What is the time dilation?
    Solution:

We have

$$\gamma = \frac{E}{m_\mu}$$

Mass of Muon $m_\mu \approx 106$ MeV

$$\gamma = \frac{20000}{106} = 188.68$$

Therefore the time dilated

$$t' = 2.6 \times 10^{-8} \times 188.68 = 4.9$$

$$\boxed{t' = 4.9\ \mu\text{s}}$$

Time is dilated by factor of $\sim$188

14. In the scattering process $a+b \to c+d$, where θ^* is the scattering angle in the CMS, show that:

$$\cos\theta^* = \frac{s(t-u) + (m_a^2 - m_b^2)(m_c^2 - m_d^2)}{\lambda^{1/2}(s, m_a^2, m_b^2)\, \lambda^{1/2}(s, m_c^2, m_d^2)}$$

And for elastic scattering, show that:

$$t = -4|\vec{p}|^2 \sin^2\left(\frac{\theta^*}{2}\right)$$

where $\vec{p}$ is the initial momentum in the CMS

Solution:

We'll use the relations

$$E_a = \frac{s + m_a^2 - m_b^2}{2\sqrt{s}} \qquad E_b = \frac{s + m_b^2 - m_a^2}{2\sqrt{s}}$$

$$|\mathbf{p}| = \frac{\lambda^{1/2}(s, m_a^2, m_b^2)}{2\sqrt{s}} \qquad |\mathbf{p}'| = \frac{\lambda^{1/2}(s, m_c^2, m_d^2)}{2\sqrt{s}}$$

$$\begin{aligned}
t &= (p_a - p_c)^2 \\
&= m_a^2 + m_c^2 - 2(E_a E_c - |\mathbf{p}||\mathbf{p}'|\cos\theta) \\
u &= (p_a - p_d)^2 \\
&= m_a^2 + m_d^2 - 2(E_a E_d + |\mathbf{p}||\mathbf{p}'|\cos\theta) \\
t - u &= m_c^2 - m_d^2 - 2E_a(E_c - E_d) + 4|\mathbf{p}||\mathbf{p}'|\cos\theta
\end{aligned}$$

$E_c - E_d = \frac{m_c^2 - m_d^2}{\sqrt{s}}$

$$t - u = m_c^2 - m_d^2 - 2\left(\frac{s + m_a^2 - m_b^2}{2\sqrt{s}}\right)\left(\frac{m_c^2 - m_d^2}{\sqrt{s}}\right) + 4\frac{\lambda^{1/2}(s, m_a^2, m_b^2)\lambda^{1/2}(s, m_c^2, m_d^2)}{4s}\cos\theta$$

$$s(t-u) = s(m_c^2 - m_d^2) - \left(s + m_a^2 - m_b^2\right)\left(m_c^2 - m_d^2\right) + \lambda^{1/2}(s, m_a^2, m_b^2)\lambda^{1/2}(s, m_c^2, m_d^2)\cos\theta$$

$$s(t-u) = -\left(s - s + m_a^2 - m_b^2\right)\left(m_c^2 - m_d^2\right) + \lambda^{1/2}(s, m_a^2, m_b^2)\lambda^{1/2}(s, m_c^2, m_d^2)\cos\theta$$

$$s(t-u) = -\left(m_a^2 - m_b^2\right)\left(m_c^2 - m_d^2\right) + \lambda^{1/2}(s, m_a^2, m_b^2)\lambda^{1/2}(s, m_c^2, m_d^2)\cos\theta$$

$$\boxed{\cos\theta = \frac{s(t-u) + (m_a^2 - m_b^2)(m_c^2 - m_d^2)}{\lambda^{1/2}(s, m_a^2, m_b^2)\lambda^{1/2}(s, m_c^2, m_d^2)}}$$

For Elastic Scattering $a = c$ and $b = d$, therefore, $|\mathbf{p}| = |\mathbf{p}'|$

$$\begin{aligned} t &= (p_a - p_c)^2 \\ &= (E_a - E_c)^2 - (\mathbf{p_a} - \mathbf{p_c})^2 \\ &= -2(|\mathbf{p}|^2 - |\mathbf{p}|^2\cos\theta) \end{aligned}$$

$$\boxed{t = -4|\mathbf{p}|^2 \sin^2\left(\frac{\theta}{2}\right)}$$

15. The decay width of the η-meson is $\Gamma_\eta = 2.6$ keV. What is the lifetime τ in seconds?

Solution:

$$\begin{aligned} \tau &= \frac{1}{\Gamma} \\ &= \frac{1}{2.6 \text{ keV}} \\ &= 0.3846 \ (\text{keV})^{-1} \end{aligned}$$

$$\begin{aligned} 1 \text{ s} &= 1.52 \times 10^{24} \ (\text{GeV})^{-1} \\ &= 1.52 \times 10^{18} \ (\text{keV})^{-1} \\ (\text{keV})^{-1} &= 6.58 \times 10^{-19} \text{ s} \\ \tau &= 0.3846 \times 6.58 \times 10^{-19} \end{aligned}$$

$$\boxed{\tau = 2.53 \times 10^{-19} \text{ s}}$$

16. The lifetime of π^0 is $\tau_{\pi^0} = 0.89 \times 10^{-16}$ s. What is the decay width Γ in natural units?

Solution:

$$0.89 \times 10^{-16}\ \mathrm{s} = 0.1353\ (\mathrm{eV})^{-1}$$

Decay Width

$$\begin{aligned}\Gamma &= \frac{1}{\tau} \\ &= \frac{1}{0.1353}\end{aligned}$$

$$\boxed{\Gamma = 7.4\ \mathrm{eV}}$$

17. Show that $d^4p = dp^0\, dp^1\, dp^2\, dp^3$ is a Lorentz-invariant quantity.
Solution:
d^4p will transform like a tensor-density in Lorentz transformation

$$d^4p' = \left| \left| \frac{\partial x'}{\partial x} \right| \right| d^4p$$

where $\left|\frac{\partial x'}{\partial x}\right|$ is the Jacobian of transformation, which is

$$\det(\Lambda^{\mu}{}_{\nu})$$

Since in a proper orthochronous Lorentz transformation $\det(\Lambda)$=1, the quantity d^4p is Lorentz invariant

18. Consider the hyperon decay $\Lambda \to p + \pi^-$. Given:

$$M_\Lambda = 1.1157\ \mathrm{GeV}, \quad M_p = 0.9383\ \mathrm{GeV}, \quad m_{\pi^-} = 0.1396\ \mathrm{GeV},$$

calculate the energy of the outgoing pion.
Solution:
We can use eqn. (3) from problem 11

$$E_\pi = \frac{M_\Lambda^2 - M_p^2 + M_\pi^2}{2M_\Lambda}$$

Therefore,

$$E_\pi = \frac{1.1157 - 0.9383 + 0.1396}{2 \times 1.1157}$$

$$\boxed{E_\pi = 0.142\ \mathrm{GeV}}$$

19. Two particles of mass m each and speed v collide at right angles forming a new body of mass M. What is the new mass M?

Solution:

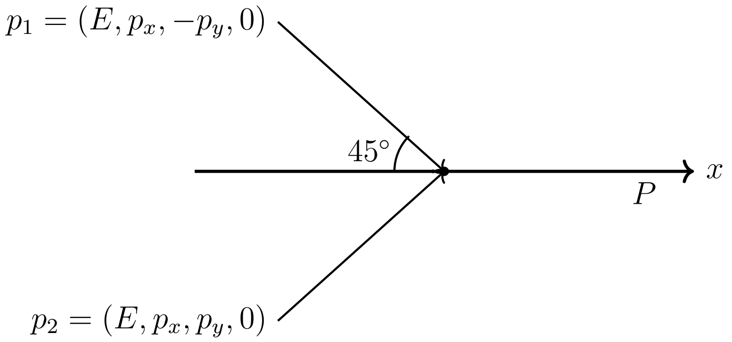

Figure 3:

Use $p^\mu p_\mu = m^2$ we have

$$\begin{aligned} p_1^2 + p_2^2 + 2p_1.p_2 &= M^2 \\ 2m^2 + 2(E^2 - p_x^2 + p_y^2) &= M^2 \\ 2m^2 + 2E^2 &= M^2 \qquad \left(\text{since } p_x = p_y = \frac{p}{\sqrt{2}}\right) \\ \frac{2m^2}{1-v^2} + 2m^2 &= M^2 \end{aligned}$$

$$\boxed{M = m\sqrt{2\left(1 - \frac{1}{1-v^2}\right)}}$$

20. Show that rapidity η is additive.

Solution:

Relativistically the velocity in same direction is added according to

$$v = \frac{v + v'}{1 + \frac{vv'}{c^2}} \tag{4}$$

Now we define rapidity ξ as

$$\beta = \frac{v}{c} = \tanh \xi$$

we can now write

$$v = c \tanh \xi$$

$$v' = c \tanh \xi'$$

Equation (4) can be written as

$$
\begin{aligned}
&= \frac{c \tanh \xi + c \tanh \xi'}{1 + \tanh \xi \tanh \xi'} \\
&= c \tanh (\xi + \xi')
\end{aligned}
$$

which shows that

$$\boxed{\xi_{\mathrm{res}} = \xi + \xi'}$$

and hence rapidity is additive